Lab 8: Telemetry System and Remote Motor Control using CAN Bus

Objective

To build a two-node CAN bus network using two Tiva C (TM4C123) boards. Node 1 reads a potentiometer, a temperature sensor and two switches and sends the data over CAN at 500 kbps. Node 2 receives the data and controls the speed and direction of two DC motors, and both nodes show the system status on a 16x2 LCD.

Components

Tiva C TM4C123GXL LaunchPad (2), EduARM4 board with 16x2 LCD (2), MCP2551 CAN module (2), potentiometer, DHT11 temperature sensor, L298N motor driver, two DC motors, 12 V supply.

Pin Connections

Signal	Node 1 (Sensor)	Node 2 (Motor)
CAN TX (to MCP2551 TX)	PE5	PE5
CAN RX (from MCP2551 RX)	PE4	PE4
LCD RS / E	PA6 / PA7	PA6 / PA7
LCD D0-D7	PB0-PB7	PB0-PB7
Potentiometer	PE3	-
DHT11 data	PD2	-
SW1 / SW2	PF4 / PF0	-
Motor-1 ENA (PWM) / IN1 / IN2	-	PF2 / PA2 / PA3
Motor-2 ENB (PWM) / IN3 / IN4	-	PF3 / PA4 / PA5

The two MCP2551 modules are connected CANH to CANH and CANL to CANL, and the grounds of both boards are connected together.

CAN Frame (ID 0x100, 5 bytes, every 100 ms)

Byte 0	Byte 1-2	Byte 3	Byte 4
Potentiometer (%)	Temperature (°C)	Switch status (motor directions)	Counter

Working

1. Node 1 reads the potentiometer using the ADC and the temperature from the DHT11.
2. Each press of SW1 or SW2 changes the direction of Motor-1 or Motor-2.
3. Node 1 sends all values in one CAN frame and shows Temperature, Potentiometer %, SW1 and SW2 status on its LCD.
4. Node 2 receives the frame. Motor-1 speed follows the potentiometer and Motor-2 speed follows the temperature, using PWM through the L298N.
5. Node 2 shows Motor-1 and Motor-2 speed (%) and direction on its LCD.

LCD Display

Node	Line 1	Line 2
Node 1	T:027°C P:045%	S1:CW S2:ACW
Node 2	M1:045% CW	M2:046% ACW

Result

The telemetry system and remote motor control over the CAN bus were implemented successfully. Motor-1 speed changed with the potentiometer, Motor-2 speed changed with temperature, and both motor directions were controlled remotely using SW1 and SW2.